



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064



A Comprehensive Study of Data Analysis, Optimization, and Simulation Techniques Using Python and SQL

AN INTERNSHIP REPORT

Submitted by

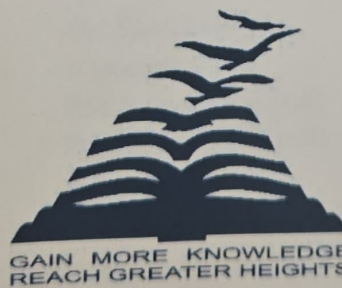
Jilpha Susmith - 20221CSD0066

Under the guidance of,

**Mr. Sahabzada Betab Badar
BACHELOR OF TECHNOLOGY**

IN

**COMPUTER SCIENCE AND ENGINEERING
(Data Science)**



**PRESIDENCY UNIVERSITY
BENGALURU
APRIL 2026**



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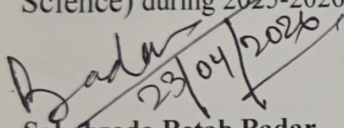
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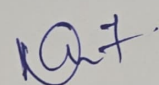


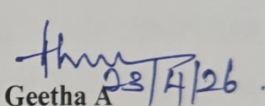
PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

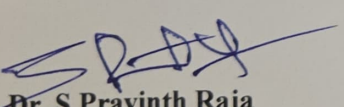
BONAFIDE CERTIFICATE

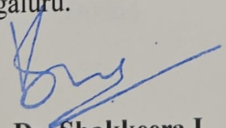
Certified that this report "A Comprehensive Study of Data Analysis, Optimization, and Simulation Techniques Using Python and SQL" is a bonafide work of **Jilpha Susmith (20221CSD0066)**, who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING (Data Science)** during 2025-2026.

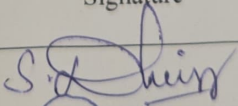
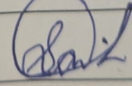

Mr. Sahabzada Betab Badar
Assistant Professor
Internship Guide
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Presidency University,
Bengaluru.


Mrs. Nisha Robin Rohit
Assistant Professor
Program Internship Coordinator
PSCS (Spl. Division)
Presidency University,
Bengaluru.


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Associate Professor
School Internship Coordinator
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Presidency University,
Bengaluru.


Dr. S Pravinth Raja
Professor & Head of the Department
PSCS (Spl. Division)
Presidency University,
Bengaluru.


Dr. Shakkeera L
Associate Dean
PSCS (Spl. Division)
Presidency University,
Bengaluru.

Sl.No	Name of the Examiner	Signature
1	Ms. Sai Dhivya S	
2	Ms. Sandhya L	



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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING DECLARATION

I am a student of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE), at Presidency University, Bengaluru, named Jilpha Susmith, hereby declare that the internship work titled “**A Comprehensive Study of Data Analysis, Optimization, and Simulation Techniques Using Python and SQL**” has been independently carried out by me and submitted in partial fulfilment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING (Data Science) during the academic year of 2025-2026. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

Jilpha Susmith

USN: 20221CSD0066

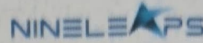
Signature

PLACE: BENGALURU

DATE: 21 - April 2026

INTERNSHIP COMPLETION CERTIFICATE

Zoho Sign Document ID: DF-B8B6C7-THURACE22AINH-LKUXVCMF9WSLD6S8LLNB1V2Q3WL9AP4Q



Intern ID: NLI-775

Date: 6th January 2026

Jilpha Susmith

Grace Haven, Sriram Apartments,
Vyasanaagar, Behind KPT, Mangalore-575004

Subject: Offer of Employment

Dear Jilpha,

Congratulations! Further to your application for employment with us, and the subsequent selection process, we are delighted to offer you a position as Junior Data Analyst with Nineleaps Technology Solutions Private Limited (the "Company"). The designation will be subject to your performance post the training period. Your location of reporting is Bangalore (this may be subject to change), India and the scheduled date of your joining will be 19th January 2026.

Offer would stand automatically canceled in case of any deviations in information or if you fail to report on or before the agreed date.

Your CTC will be Rs. 6,60,000/- per annum (Six Lakh Sixty Thousand Rupees Only). This CTC includes the fixed and variable component as detailed in the Compensation Details sheet provided under **Annexure**

Your appointment will be governed by the terms and conditions of employment that you shall sign at the time of joining. Any breach of the same may result in withdrawal of this offer and or termination of your employment.

Please sign the duplicate of this offer as your acceptance and forward the same to us. If you do not do so within seven days after the date of this letter, the offer will lapse.

Welcome to the Nineleaps family! We look forward to your joining us and wish you a long and fruitful career with the Company.

Yours Sincerely,

for Nineleaps Technology Solutions Private Ltd.

Abhinaya K
Human Resource

I have read, understood, and accept the terms and conditions and will join on Jan 19 2026 12:12

Signature: *Jilpha Susmith*

Date: Jan 19 2026 12:12

This is a computer generated letter. Signature not required.

3025 & 3030, 80 Feet Road, 1st Block Konomangala, Bangalore, Karnataka 560034

ANNEXURE A

The terms & conditions of our offer are as follows:

1. Training Period:

You will undergo a training period of 6 months from the date of joining the Company. This training period may be extended based on your performance and at the discretion of the Company. Post the successful completion of this training period, you will be placed on Probation. You will be entitled to a monthly stipend of ₹ 20,000/- during the training period.

2. Probationary Period:

You will be on probation for a period of 3 months from the date of completion of your training period and may be confirmed as a permanent employee upon successful completion of your probation. If the Company does not find your performance satisfactory, it reserves the right to not proceed with employment.

3. Background Check:

Your confirmation of employment is subject to you submitting the requisite documents as required by the Company and successful completion of background checks to validate your identity. You undertake to provide authentic, true, and complete information and provide your consent to use the information for the purpose of conducting the background check. If the outcome of the background check is not desirable, the Company reserves the right to withdraw this Offer without notice and Compensation and/or termination of your employment.

4. Notice Period:

You will be required to give a 3 month notice or salary thereof in case you decide to leave our services subject to the Company's discretion. In the event of you having any incomplete assignment, the Company will have the discretion to relieve you only at the end of the three-month notice period. Similarly, the Company can terminate your services by giving 3 months' notice or fixed salary thereof. For reasons of clarity, during your probation period, you are required to give a notice of 3 months. Similarly, the Company can terminate your services by giving you a notice of 3 months, during your probation period.

5. Confidential Information:

You will not divulge or communicate to any other person or persons, natural or juristic, directly or indirectly, any information about the affairs of the Company or the management at any time or any other information deemed confidential to the Company. This includes but not limited to any Intellectual Property (IP) of the Company, or any IP developed by you, during your term at the Company.

6. Other Terms & Conditions:

You agree not to undertake employment, whether full-time or part-time, as the Director / Partner / Member / Employee of any other organization / entity engaged in any form of business activity without the consent of the Company and in accordance with Company policies.

Jilpha Gusmith

This Jan 19 2026 12:12 IST signed

2225 & 2220, 80 Feet Road, 1st Block Kormangala, Bengaluru, Karnataka 560034

7. Non-Compete:

You agree that following the termination of your employment with the Company for any reason, you will not accept any offer of employment for a period of 6 months; from any of the Company's customers with whom you would have worked in a professional capacity during your term at the Company.

8. Company Policy:

You are required to adhere to the Company's policies and procedures (subject to change from time to time) that are detailed in the employee handbook, which will be handed over to you at the time of joining.

ANNEXURE B
Compensation Details

(All figures in Indian Rupees per year unless specified otherwise)

Compensation Structure

Component	Yearly	
Basic:		270,000
Allowances:		
a) House Rent Allowance	108,000	
b) Special Allowance	184,110	292,110
Retirals & Other Benefits:		
a) Employer Contribution of PF	21,600	
b) Gratuity	12,981	
c) Insurance - GMC (Health Insurance) Self, Spouse & Child, GPA (Accidental Insurance)	3,309	37,890
Annual Variable Pay:		60,000
Total Cost to Company (Basic + Allowance + Retirals + Variable Pay)		660,000

* The above CTC will have deductions with respect to Income tax, Provident Fund, Professional Tax, and Gratuity.

The Annual Variable Incentive will be paid on a yearly basis at the end of the financial year (March). The variable incentive is calculated based on the employee's utilization status, i.e., the number of days billable on a project. Employees who have resigned and are serving their notice period will not be

All the above benefits are as per Company's policies, which are subject to change from time to time. You shall be duly notified in the event of any such change. Please note that there will be statutory deductions of income tax, professional tax and PF out of the gross salary.

Jilpha Gasmith

This is a computer-generated letter. Signature not required.

3025 & 1030, 80 Feet Jan 19 2026 12:12 IST

ACKNOWLEDGEMENT

For completing this internship work, I have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. I extend my gratitude to our beloved Chancellor, Pro-Vice Chancellor, and Registrar for their support and encouragement in completion of the internship.

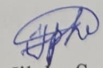
I would like to sincerely thank my internal guide Mr. Sahabzada Betab Badar, Assistant Professor, Presidency School of Computer Science and Engineering (Spl. Division) for his moral support, motivation, timely guidance and encouragement provided to me during the period of internship work.

I am also thankful to Dr. S Pravinth Raja, Professor, Head of the Department, Presidency School of Computer Science and Engineering (Spl. Division) for his mentorship and encouragement.

I express my cordial thanks to Dr. Shakkeera L, Associate Dean, Presidency School of Computer Science and Engineering (Spl. Division) and the Management of Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my internship work.

I am grateful to Dr. Geetha A, PSCS (Spl. Division) School Internship Coordinator, and Mrs. Nisha Robin Rohit, Program internship coordinator Presidency School of Computer Science and Engineering for facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance.

I am also grateful to Teaching and Non-Teaching staff of Presidency School of Computer Science and Engineering and also staff from other departments who have extended their valuable help and cooperation.



Jilpha Susmith

ABSTRACT

This internship was undertaken with the objective of gaining practical exposure to real world applications of data science and analytics. In today's data driven environment, organizations rely heavily on data to make informed decisions and improve performance. The internship provided an opportunity to work on multiple projects across different domains, including sports analytics, simulation techniques, database management, and programming. Through these projects, the focus was on understanding how theoretical knowledge can be applied to solve real problems effectively.

The approach followed during the internship was structured and systematic, beginning with problem understanding and data collection, followed by data preprocessing, analysis, and modeling. Various tools and technologies such as Python, SQL, and optimization techniques were used to implement solutions. In the IPL team optimization project, a data driven model was developed to select the best possible team under given constraints. The Monte Carlo simulation project focused on analyzing uncertainty and variability by generating multiple possible outcomes. SQL assignments were used to manage and query structured data efficiently, while Python based projects involved data exploration and the development of a customer management system. Each project followed a logical workflow, ensuring that the results were accurate, reliable, and meaningful.

The results of the internship demonstrate the effectiveness of applying analytical techniques to real world problems. The IPL project successfully identified an optimal team composition, highlighting the importance of data driven decision making. The Monte Carlo simulation provided insights into risk and uncertainty, showing how probabilistic methods can improve understanding of complex systems. SQL and Python projects reinforced the importance of data management and preprocessing in achieving accurate outcomes. Overall, the internship enhanced technical skills, analytical thinking, and problem solving abilities, providing a strong foundation for future work in the field of data science and analytics.